

Hands-on Lab 01 — Cybersecurity Fundamentals

Course: 420-D01-AB · *Preventive Monitoring in Cybersecurity*

Cohort: CYB-01

Day: 02

Estimated student workload: 5 hours of *work* (not of reading)

Maximum deliverable: 4 A4 pages, PDF/A only

Due: Sunday, 17 May 2026, 23:59 EST (Eastern Standard Time, UTC-5)

Late penalty: 10% per calendar day

Weight in final grade: 5%

0. Ownership & instructor protection notice — read this first

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You may **not** redistribute this brief, repost it on file-sharing portals, push it into a public AI training corpus, or hand it to peers outside cohort CYB-01. Internal study with classmates is fine; publication is not. Breach of this clause will be treated as a violation of academic integrity and pursued under the cheating-and-plagiarism procedure cited in your course outline.

1. Legal and ethical disclaimer — read it twice

Network monitoring, packet inspection, log triage and traffic-pattern analysis use techniques that are functionally identical to those used in offensive security operations. The fact that you are doing this for course credit does **not** legalise running these techniques against networks you do not own or for which you do not hold explicit written authorisation.

You shall **not**:

- Capture or analyse traffic from any network you do not own or have permission to monitor — that includes home routers shared with roommates, café Wi-Fi, employer networks, and any educational institution's infrastructure.
- Re-host, share, or repackage the malware samples or simulated payloads embedded in any dataset you download for this lab. Hash references are fine; binaries are not.
- Use the techniques covered here against any production system, anywhere, regardless of how curious you are.
- Attempt to identify, contact, or de-anonymise any person, organisation, or victim whose data may appear in a public sample dataset.

Applicable legal frameworks include — and are not limited to:

- [Canadian Criminal Code, Section 342.1 — Unauthorized Use of Computer](#)
- [United States — Computer Fraud and Abuse Act, 18 U.S.C. § 1030](#)
- [European Union Directive 2013/40/EU on attacks against information systems](#)
- [Spain — Ley Orgánica 10/1995, del Código Penal \(Articles 197 bis, 197 ter and 264\)](#)

Cyber.SoHo and the instructor delivering this session bear no liability for any misuse of the techniques presented. By submitting your lab report you confirm — implicitly through the honour declaration in §9 — that all work was performed inside an isolated, authorised laboratory environment.

2. Tooling notice — everything below is a suggestion, nothing is mandatory

Every tool, dataset, framework and method named in this document is a **suggestion**. The picks were made because they are free, open-source, well-documented, platform-agnostic, and battle-tested by real practitioners. That is the only reason. None of them is required.

If you have a packet analyser you already prefer, a SIEM (Security Information and Event Management) you have used at work, a log parser of your own creation, or you want to write a fifty-line Python script that does the job better — go ahead. We strongly encourage you to:

- **Research alternatives before defaulting to the first tool listed.** A two-hour evaluation now saves you weeks during your career.
- **Document why you picked the tool you used.** A defensible methodology is half the report.
- **Build your own tool if nothing fits.** A custom Python script with `scapy`, `dpkt`, or `pyshark` is a perfectly valid submission and will be graded favourably for originality.
- **Mix-and-match.** Wireshark for triage, a Python notebook for stats, CyberChef for decoding — that is the real workflow.

This is a cybersecurity course. Curiosity, lateral thinking and the willingness to step outside a vendor's marketing pitch are core competencies, not optional extras.

3. Learning outcomes

By the end of this five-hour lab you will be able to:

1. Acquire a simulated network capture and a corresponding set of operating-system logs from a reputable open-data source, hash-verify them, and load them into an isolated analysis environment.
2. Inspect packet captures with a network protocol analyser and surface indicators of suspicious behaviour without prior knowledge of the underlying scenario.
3. Apply at least three threat-classification frameworks — CIA (Confidentiality, Integrity, Availability) triad, STRIDE, MITRE ATT&CK (Adversarial Tactics, Techniques, and Common Knowledge) — to your findings.
4. Conduct a structured, traceable, documented risk assessment on a fictional small organisation using a recognised methodology.
5. Produce a four-page research report that holds up to academic scrutiny on structure, citation, attribution, and originality.

These outcomes map to course competency **EZ01**, Elements 1, 2, and (partially) 4.

4. Time budget

Block	Activity	Allocated time
§5	Environment setup	30 min
§6.2	Part A — Traffic capture analysis	90 min
§6.3	Part A — Log analysis	60 min
§7	Part B — Risk assessment	75 min
§8	Report drafting and polish	45 min
	Total	5 h 00 min

These are working estimates. If you finish a block early, use the time to deepen the next one rather than ending early — depth is what separates a competent submission from a memorable one.

5. Block 1 — Environment setup (30 min)

5.1 Workstation baseline

You need a machine — physical or virtualised — with:

- 8 GB RAM minimum, 16 GB recommended
- 30 GB free disk
- Internet access only during dataset download; isolate (NAT-only or air-gap) for analysis

5.2 Suggested base operating system (pick one, swap freely)

- [Ubuntu 24.04 LTS Desktop](#) — clean, predictable, free
- [Kali Linux 2025.x](#) — most analysis tools pre-installed
- [Parrot Security OS](#) — lighter Kali alternative
- [Microsoft Windows 11 Pro](#) — works fine, you install the toolkit yourself

A virtual machine is strongly preferred over running on bare metal. [Oracle VirtualBox](#) (free) and [VMware Workstation Pro](#) (free for personal use as of 2024) both work.

5.3 Suggested analysis toolkit

Purpose	Suggested tool	Reasonable alternatives
Packet analysis	Wireshark	tshark, tcpdump, Brim/Zui, Arkime
Log parsing	grep, awk, jq, Python	Logstash, Splunk Free, Graylog Open
Hash verification	sha256sum, certutil, Get-FileHash	hashdeep, rhash
Hash & IOC lookup	VirusTotal	AbuseIPDB , AlienVault OTX
Malware sample reference (read-only)	MalwareBazaar	URLhaus
MITRE ATT&CK mapping	ATT&CK Navigator	manual STIX export, DeTT&CT
Universal data transformation	CyberChef	local Python scripts

Swap any of these for an equivalent you prefer. State the swap and your reason in your report's methodology section.

5.4 Hash-and-verify everything you download

Before opening any file from any source listed in §6.1 — public or not, "trusted" or not — verify the SHA-256 hash. Always.

```
# Linux / macOS
sha256sum suspicious_capture.pcap

# Windows PowerShell
Get-FileHash -Algorithm SHA256 .\suspicious_capture.pcap
```

If the hash does not match the publisher's stated value: stop, do not proceed, report it through your course channel.

6. Block 2 — Part A: Threat identification & classification (2 h 30 min)

6.1 Choose your simulated dataset

Pick **one** packet capture and **one** log set from the lists below. You may also propose your own dataset; describe its provenance, licence and integrity check in your report.

Open packet capture sources (PCAP — Packet Capture, in libpcap format):

- [Malware-Traffic-Analysis.net](#) — Brad Duncan's archive of real-world infection traffic exercises. Pick any "exercise" dated 2023–2025. Each entry comes with a write-up — **do not read the write-up before you finish your own analysis**.
- [The Honeynet Project Challenges](#) — captures with redacted answers; ideal pedagogical fit.
- [Wireshark Sample Captures wiki](#) — protocol-level samples; good for the "calm" half of your analysis.
- [CIRA-CIC-DoHBrw-2020](#) — Canadian Institute for Cybersecurity, DNS-over-HTTPS dataset, labelled benign/malicious flows.
- [Stratosphere IPS — CTU-13 botnet captures](#) — captures from CTU University, Prague.

Open log sources:

- [SecRepo Samples Repository](#) — Mike Sconzo's collection of security-relevant logs (Apache access, Bro/Zeek, Windows Event, Snort, etc.)
- [Splunk Boss of the SOC v3 \(BOTSV3\) public dataset](#) — Windows Event, Sysmon, Stream, Suricata, full SIEM corpus.

- [LANL Comprehensive Cyber-Security Events](#) — Los Alamos National Laboratory anonymised authentication logs.
- [Aktaion Data](#) — labelled exploit kit traffic + Bro logs.

A note on the wider question of "do these datasets have permission for educational use" — yes, each of the sources above publishes its terms; read them before downloading. Cyber.SoHo did not produce, modify, or redistribute any third-party dataset. You are responsible for compliance with the dataset publisher's licence.

6.2 Step-by-step traffic analysis (90 min)

Stop before you proceed. Confirm you are inside the isolated VM. The PCAP itself is inert as a file, but it may contain URLs or DNS names that some preview tools will silently resolve. Disable your network adapter or set the VM to host-only before you open the file.

Step 1 — Load the PCAP.

File → Open → suspicious_capture.pcap

Step 2 — Generate a protocol hierarchy report.

Statistics → Protocol Hierarchy

Note the percentage of bytes on each protocol. Anything that does not match your expectation — a 40% DNS share, unsolicited SMB chatter, raw TCP streams to high non-standard ports — becomes your first lead. Take a screenshot. You will use it in your report.

Step 3 — Generate the conversations table.

Statistics → Conversations → IPv4 / IPv6 / TCP / UDP tabs

Sort by bytes, then by packets. The chattiest pair of endpoints is rarely the legitimate one in a compromised environment.

Step 4 — Run a DNS breakdown.

Statistics → DNS

Hunt for: very long domain names, high-entropy random-looking strings (a hallmark of DGA — Domain Generation Algorithm), DNS over non-standard ports, query names that resolve to the same IP repeatedly.

Step 5 — Build your own display-filter cheat-sheet. Start from these and extend the list as you find your own:

```
http.request and !(http.host contains "google" or http.host contains "microsoft")
```

```
tcp.port == 4444 or tcp.port == 8080 or tcp.port == 1337
dns.qry.name matches "[a-z0-9]{20,}"
tls.handshake.extensions_server_name contains ".onion"
ip.dst in {a.b.c.d, e.f.g.h}    # paste suspicious destinations
```

Save your cheat-sheet — three to five filters of your own composition is a graded artefact in your report.

Step 6 — Follow two suspicious streams.

Right-click packet → Follow → TCP Stream

Dump the payload. Compute its SHA-256. Submit the **hash** (never the payload) to [VirusTotal hash search](#). Capture the result. If the dataset's licence explicitly permits sample re-upload, you may submit the file itself; otherwise, hash only.

Step 7 — Map your findings to the MITRE ATT&CK matrix using the [ATT&CK Navigator](#). Tag each technique with confidence (low / medium / high). Export the layer as a screenshot for your report.

6.3 Step-by-step log analysis (60 min)

Step 1 — Inventory the log set. How many files, what formats, what time range? Note the absolute earliest and latest timestamp — that is your "incident window".

Step 2 — Establish a baseline for one calm hour. Useful baseline metrics:

- Authentication failures per minute
- HTTP status code distribution
- DNS queries per source IP
- Sysmon Event ID 1 (process creation) per host

Step 3 — Hunt for outliers. Three worked starting patterns:

```
# Brute-force candidates in an OpenSSH auth.log
grep "Failed password" auth.log | awk '{print $11}' | sort | uniq -c | sort -rn | head -20
```

```
# Suspicious User-Agent strings in an Apache access.log
awk -F'"' '{print $6}' access.log | sort | uniq -c | sort -rn | head -20
```

```
# Sysmon process creation events with command-line length over 200 chars
jq 'select(.EventID==1 and (.CommandLine|length)>200) | {Computer, Image, CommandLine}' sysmon.json
```

Step 4 — Cross-reference packet findings against log findings. Source IPs that show up in both data sources are your first corroborated indicators. This crossover is the most valuable evidence you will produce in this lab.

6.4 The Part A deliverable — one consolidated classification table

Produce a table with the following columns. **Five rows minimum, ten rows maximum.**

#	Indicator (sanitised)	Source (PCAP / log)	CIA impact (C / I / A)	STRIDE category	MITRE ATT&CK Tactic + Technique ID	Confidence (Low / Med / High)
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A note on sanitisation. Replace public IPv4 addresses with their first two octets followed by .x.x (e.g., 185.199.x.x). Redact usernames. Mask the dataset's victim hostnames. This protects any third party whose data may be embedded in a public sample.

A reminder on the three frameworks — expanded for your reference:

- **CIA (Confidentiality, Integrity, Availability)** — what does the indicator threaten?
 - **STRIDE (Spoofing, Tampering, Repudiation, Information disclosure, Denial of service, Elevation of privilege)** — Microsoft's threat-modelling taxonomy.
 - **MITRE ATT&CK (Adversarial Tactics, Techniques, and Common Knowledge)** — adversary-behaviour knowledge base maintained by [The MITRE Corporation](#).
-

7. Block 3 — Part B: Risk assessment mini exercise (75 min)

7.1 Methodology — pick one and stick with it

Pick a recognised methodology and use it consistently throughout the exercise. Do not blend.

- [NIST SP 800-30 Revision 1 — Guide for Conducting Risk Assessments](#) — qualitative, very explicit step list, recommended for first-timers.
- [ISO/IEC 27005:2022 — Information security risk management](#) — paywalled standard; the structure is widely summarised in free derivative works.
- [FAIR — Factor Analysis of Information Risk](#) — quantitative, more demanding, more rewarding if you have a statistics background.

State your choice in your report and defend it in one sentence.

7.2 Scenario — Northwind Apothecary Ltd. (fictional)

Northwind Apothecary Ltd. is a fictional 22-employee specialty pharmacy operating from a single physical location in downtown Halifax, Nova Scotia. Its information assets:

- Customer prescription database, **PostgreSQL 16**, approximately 14,000 records, regulated under provincial health-information privacy law.
- Point-of-Sale (POS) workstations: four Windows 11 Pro machines.

- Inventory-management server: Debian 12, on-premises.
- Wi-Fi guest network for waiting customers, open SSID, captive portal.
- Wi-Fi staff network: SSID "NW-Staff", WPA2-PSK, shared password rotated yearly.
- One ageing Cisco SOHO (Small Office / Home Office) router; default admin port still on 80/HTTP; firmware four years old.
- Cloud backup via a free-tier consumer service, single shared owner account, no MFA (Multi-Factor Authentication).
- One part-time IT contractor, on-site twice a month.
- The general manager works from home one day a week using their personal laptop and the staff Wi-Fi password.

Last quarter, an employee clicked on a phishing email that purported to be from a wholesale supplier; nothing visible happened, but the IT contractor flagged it during a routine review.

Northwind Apothecary Ltd. is fictional. Any resemblance to a real pharmacy, anywhere, is coincidental.

7.3 Required steps

Step 1 — Asset register. List every asset visible in the scenario. Rate each one Low / Medium / High / Critical for criticality, and justify the rating in one line.

Step 2 — Threat sources. Identify at least six. Cover all four NIST 800-30 categories: *adversarial, accidental, structural, environmental*. A "structural" threat is, for example, the consumer-grade router's firmware support reaching end-of-life. An "environmental" threat is, for example, a flood in downtown Halifax. Both belong on your list.

Step 3 — Vulnerability mapping. Map each asset to the specific weaknesses visible in the scenario. A vulnerability is not "the asset is bad"; it is a concrete, named weakness — *"router admin interface on cleartext HTTP, accessible from staff Wi-Fi"*, not *"the router is old"*.

Step 4 — Likelihood and impact rubric. Use a 1–5 scale for each. Document your rubric. A one-paragraph definition is fine, but you must define what each level means. Example for likelihood:

1 — Unlikely in a five-year window. 5 — Expected within the next quarter.

Step 5 — Risk register. One row per asset / threat / vulnerability triple:

ID Asset Threat Vulnerability L (1-5) I (1-5) Risk score (L × I) Recommended treatment

Treatment must be one of: *Mitigate, Transfer, Accept, Avoid*. Justify your choice in three to ten words.

Step 6 — Top three priorities. From your full register, surface the three highest-scoring risks. Write a two-sentence justification for each. This is the section your hypothetical executive sponsor would read first; write it so they actually would.

8. Block 4 — The four-page research report (45 min)

This is the deliverable that gets graded. **Four A4 pages maximum.** Times New Roman or a reasonable equivalent at 11 pt, single-spaced, 2 cm margins. Going over costs you marks. Coming in lean — say three and a half pages — is rewarded.

8.1 Required structure

Page 1 — Cover and abstract

- Title, your name, student ID, course code 420-D01-AB, cohort CYB-01, date of submission.
- Cyber.SoHo wordmark — provided in the cohort's shared folder.
- 150-word abstract: scenario, methods, key findings, one-line conclusion.

Page 2 — Methodology

- Tools and dataset selected, with one-sentence justification each.
- Methodology framework chosen for the risk assessment, with the one-sentence defence.
- Environment description (operating system, virtual machine type, network isolation method).

Pages 2–3 — Findings (Part A)

- The classification table.
- Two annotated screenshots — one of Wireshark output, one of your MITRE ATT&CK Navigator layer.
- Three to five sentences interpreting what the table tells you. *Do not* describe the table; explain its implications.

Pages 3–4 — Risk assessment (Part B)

- Asset register, compact format.
- Risk register; show the top five rows in the body, full register in an appendix if you exceed five.
- Three-paragraph executive summary of the top three risks.

Page 4 — Conclusion and references

- Three-sentence reflection on what surprised you in the data.

- Two-sentence note on what you would do differently with more time.
- Reference list, APA 7th edition. See §8.4.

8.2 Screenshots — how to take them properly

- Crop tightly. Empty desktop chrome is wasted page real-estate.
- Resolution at least 96 DPI for screen, 150 DPI if you anticipate it being printed.
- Caption every figure: *Figure 3 — Wireshark protocol hierarchy, 14:32–15:01 UTC*.
- Redact in-frame: usernames, machine names that are not fictional, any internal IPv4 that is not in RFC 1918 ranges (10.0.0.0/8, 172.16.0.0/12, 192.168.0.0/16).
- Save as PNG. JPEG compression artefacts on text screenshots look unprofessional.
- Do not rely on the operating system's default screenshot tool if it watermarks. Verify the saved file before embedding.

8.3 Images and intellectual-property protection

If you embed any image you did not produce yourself:

- Source it from a known-clean pool: [Wikimedia Commons](#), [Unsplash](#), [Pexels](#), or a vendor's official press kit.
- Verify the licence. Acceptable: Creative Commons CC-BY, CC-BY-SA, CC0, public domain, vendor "press use". Not acceptable: anything else.
- Caption with: *Source: <organisation> — <licence> — <URL>*.
- Do **not** screenshot copyrighted material from textbooks, academic papers, or competitor courseware. That includes slides from earlier lectures unless you took the screenshot yourself in real time.
- Logos and trademarks: only use the official asset from the vendor's brand-kit page; never recreate, never skew, never recolour.

8.4 Citations and references

Use **APA 7th edition** unless your discipline traditionally requires another style — in which case use what your discipline expects, consistently. Every external source, dataset, tool documentation page, framework you relied on, or piece of supporting evidence must be referenced. Inline as (Author, Year). Reference list on the last page.

URL hygiene:

- Every URL in your report must be reachable on the day of submission.
- Submit each unique URL through [VirusTotal URL scan](#) and confirm a clean result. Save the screenshot of the verification — you may be asked for it during grading.
- Use the official URL for every organisation cited. `microsoft.com`, never `microsoft-secure.cn`.

Reliable starting points for cited frameworks and bodies:

- [National Institute of Standards and Technology \(NIST\)](https://nist.gov) — nist.gov
- [International Organization for Standardization \(ISO\)](https://iso.org) — iso.org
- [Center for Internet Security \(CIS\)](https://cisecurity.org) — cisecurity.org
- [The MITRE Corporation — ATT&CK](https://attack.mitre.org) — attack.mitre.org
- [Office of the Privacy Commissioner of Canada \(OPC\)](https://priv.gc.ca) — priv.gc.ca
- [Information and Privacy Commissioner of Ontario \(IPC\)](https://ipc.on.ca) — ipc.on.ca

8.5 Plagiarism — zero tolerance

Direct quotes are limited to fifteen words and must appear inside quotation marks with an inline citation. Paraphrasing without citation is plagiarism. Submitting AI-generated prose as your own writing is plagiarism — you may use an AI assistant to brainstorm or to draft a single sentence you then rework, but you may not paste an AI-written paragraph and present it as your own. Every submission is run through both an originality checker and an AI-detection tool. The instructor reserves the right to defer grading pending a five-minute oral verification — be prepared to walk through your own report, live, on demand.

8.6 Hyperlinks

Every hyperlink you include in your report must:

- Point to the official URL.
- Be visible — do not hide URLs behind opaque "click here" labels. Link the descriptive text instead: *the [NIST risk-assessment guide](#) recommends...*
- Be styled distinctly — underlined, in blue at minimum.

9. Submission

- **Filename:** LAB01_<LASTNAME>_<FIRSTINITIAL>_420-D01-AB.pdf
Example: LAB01_TREMBLAY_M_420-D01-AB.pdf
- **Format:** PDF/A only. No .docx, no Google Docs links, no Pages exports, no zip archives.
- **Channel:** the cohort's course platform — Microsoft Teams or Moodle assignment dropbox, as instructed.
- **Deadline:** Sunday, 17 May 2026, 23:59 EST (UTC-5).
- **Late policy:** 10% deduction per calendar day, per the course outline.

Append a short honour declaration on the last page:

I confirm that the work submitted is my own. Where I have used external sources, I have cited them. I have not used any automated tool to write substantial portions of this text.

— <Your full name>, <Date>

10. Grading rubric (out of 5%)

Criterion	Weight
Technical accuracy of threat identification	25%
Quality of classification across CIA, STRIDE and MITRE ATT&CK	20%
Risk assessment rigour and traceability	25%
Report structure, clarity and professional presentation	15%
Citations, hyperlinks and image attribution	10%
Reflection and originality of insight	5%

Appendix A — Open tool catalogue (further suggestions)

- [Wireshark](#) — packet analyser
- [Brim / Zui](#) — Zeek-based PCAP triage with a SQL-like query layer
- [Zeek \(formerly Bro\)](#) — network security monitor
- [Suricata](#) — open IDS / IPS (Intrusion Detection / Prevention System)
- [Security Onion](#) — pre-built monitoring distribution
- [Splunk Free](#) — 500 MB/day cap, otherwise full SIEM
- [Graylog Open](#) — free SIEM-lite
- [Elastic Stack — Elasticsearch, Logstash, Kibana](#) — community edition
- [CyberChef](#) — universal data transformer
- [Sysmon \(System Monitor\)](#) — Windows event-log enrichment
- [Velociraptor](#) — endpoint visibility and digital forensics
- [osquery](#) — host-state queries through SQL syntax

Appendix B — Open dataset catalogue (further suggestions)

- [CICIDS2017](#) — Canadian Institute for Cybersecurity intrusion detection dataset
- [CIRA-CIC-DoHBrw-2020](#) — DNS-over-HTTPS dataset
- [CTU-13](#) — Stratosphere IPS botnet captures
- [Kyoto 2006+](#) — honeypot traffic, Kyoto University

Appendix C — Glossary of expanded abbreviations

- **APA (American Psychological Association)** — citation style. apastyle.apa.org
- **ATT&CK (Adversarial Tactics, Techniques, and Common Knowledge)** — see MITRE.
- **CFAA (Computer Fraud and Abuse Act)** — US federal computer-crime statute.
- **CIA (Confidentiality, Integrity, Availability)** — foundational security triad.
- **CIS (Center for Internet Security)** — non-profit publishing the CIS Controls.

- **DGA (Domain Generation Algorithm)** — malware technique to create rendezvous domains on the fly.
- **DNS (Domain Name System)** — name-to-IP resolution protocol.
- **DPI (Dots Per Inch)** — image resolution.
- **EST (Eastern Standard Time)** — UTC-5, used in your submission deadline.
- **IDS / IPS (Intrusion Detection / Prevention System).**
- **IOC (Indicator of Compromise)** — observable pointing to a possible incident.
- **IP (Internet Protocol)** — and IPv4 / IPv6 are its versions.
- **ISO (International Organization for Standardization)** — global standards body.
- **MFA (Multi-Factor Authentication).**
- **MITRE** — non-profit US research organisation; maintains ATT&CK.
- **NIST (National Institute of Standards and Technology)** — US standards body.
- **PCAP (Packet Capture)** — file format, libpcap.
- **POS (Point of Sale).**
- **RFC (Request for Comments)** — Internet Engineering Task Force standards series.
- **SHA-256 (Secure Hash Algorithm, 256-bit)** — cryptographic hash function.
- **SIEM (Security Information and Event Management)** — log aggregation and correlation.
- **SOHO (Small Office / Home Office)** — also our brand name; the resemblance is intentional.
- **SSID (Service Set Identifier)** — Wi-Fi network name.
- **STIX (Structured Threat Information eXpression)** — threat-intelligence interchange format.
- **STRIDE** — Microsoft threat-modelling taxonomy (see §6.4).
- **TCP / UDP (Transmission Control Protocol / User Datagram Protocol).**
- **UTC (Coordinated Universal Time)** — global time reference.
- **VM (Virtual Machine).**
- **WPA2-PSK (Wi-Fi Protected Access 2 — Pre-Shared Key).**

Appendix D — How this lab fits into your full project

Lab 01 is the first of six hands-on labs that — together with the Course Procedure Manual due on Day 14 — form the integrative project for course 420-D01-AB. The remaining labs:

- **Lab 02 (Day 04)** — Map a network architecture and identify vulnerabilities.
- **Lab 03 (Day 06)** — Configure Wireshark, SNMP and Splunk on Windows and Linux.
- **Lab 04 (Day 09)** — Traffic and log analysis: IDS alerts, rogue access, etc.
- **Lab 05 (Days 10–11)** — Live threat detection, attack simulation and incident handling.
- **Lab 06 (Days 12–13)** — Auto-analysis scripting, performance impact, cloud-based monitoring with [AWS CloudWatch](#) and / or [Microsoft Azure Sentinel](#).

Plan for cumulative growth. Findings and methods from Lab 01 should be referenceable in Lab 06. Keep your data, keep your hashes, keep your notes.

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